Question	Answer	Mark
3(a)	$v = u + at$ $v = 9.6 - (9.81 \times 0.37) = 6.0\,\text{m s}^{-1}$	A1
3(b)	$s = ½ \times (9.6 + 6.0) \times 0.37$ **or** $6.0^2 = 9.6^2 - (2 \times 9.81 \times s)$ **or** $s = (9.6 \times 0.37) - (½ \times 9.81 \times 0.37^2)$ **or** $s = (6.0 \times 0.37) + (½ \times 9.81 \times 0.37^2)$	C1
	$s = 2.9\,\text{m}$	A1
3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1
	$\Delta E = 0.056 \times 9.81 \times 2.9$ $= 1.6\,\text{J}$	A1
3(c)(ii)	$E = ½mv^2$	C1
	$\Delta E = ½ \times 0.056 \times (6.0^2 - 3.8^2)$ $= 0.60\,\text{J}$	A1
3(d)	force on ball (by ceiling) <u>equal</u> to force on ceiling (by ball)	M1
	and opposite (in direction)	A1
3(e)	$(p =)\ mv$ **or** 0.056×6.0 **or** 0.056×3.8	C1
	change in momentum $= 0.056 \times (6.0 + 3.8)$ $= 0.55\,\text{N s}$	A1

Question	Answer	Mark
3(f)	resultant force = 0.55 / 0.085 (= 6.47 N)	C1
	force by ceiling = 6.47 – (0.056 × 9.81) = 5.9 N	A1

Question	Answer	Mark
4(a)	(Young modulus =) stress / strain	B1